

MEDTRACK DV

HOSPITAL OPERATIONS & PATIENT ANALYTICS

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TOOLS USED: Kaggle • Python • Jupyter Notebook • Pandas • Tableau • GitHub

PROJECT WORKFLOW: Data Collection → Data Cleaning → Data Refinement → Data Visualization → Dashboard Development → Insights

1.INTRODUCTION

1.1 Project Background

Hospitals generate large amounts of data through patient admissions, discharges, departments, wards, beds, diseases, billing, prescriptions, diagnostic tests, and payment activities. Analysing this information manually can make it difficult to identify operational patterns and understand how hospital resources are being utilized.

The MedTrack DV – Hospital Operations & Patient Analytics project was developed to transform hospital data into meaningful and interactive visual insights. The project focuses on analysing patient flow, departmental performance, hospital resources, bed utilization, billing, and other operational indicators through dashboards.

1.2 Problem Statement

Raw hospital datasets contain detailed information but are not always easy to interpret directly. Without proper data cleaning and visualization, important patterns such as admission trends, patient movement, length of stay, department workload, bed utilization, and payment behaviour can be difficult to identify.

Therefore, the project aims to **refine the raw hospital dataset and develop interactive Tableau dashboards that present important hospital performance indicators in a clear and understandable form.**

1.3 Project Objectives

The major objectives of the project are:

- To collect a suitable hospital dataset from **Kaggle** for analysis.
- To understand the structure and quality of the collected dataset.
- To clean and refine the raw data using **Python and Pandas in Jupyter Notebook**.
- To identify and prepare relevant fields required for analysis.
- To develop meaningful **KPIs, charts, and interactive visualizations**.
- To create four Tableau dashboards covering hospital overview, patient flow, department analytics, and resource utilization.
- To provide an interactive way to explore hospital operations using filters and dashboard navigation.
- To derive meaningful insights from the available hospital data.

1.4 Project Workflow

The complete project follows a structured data analytics workflow:

Kaggle Dataset → Data Understanding → Data Cleaning & Refinement in Jupyter Notebook → Cleaned Dataset → Tableau Visualization → Dashboard Development → Analysis & Insights

2.DATA COLLECTION

2.1 Data Source

The hospital dataset was collected from **Kaggle**, an online platform that provides datasets for data analysis and machine learning projects. The dataset was selected because it contains information relevant to hospital operations and patient management.

2.2 Dataset Description

The dataset contains 45,000 hospital admission records with information related to:

- Patient admissions and discharges
- Patient demographics
- Departments and wards
- Bed availability and status
- Diseases and disease categories
- Length of stay
- Billing and payment details
- Prescriptions and diagnostic tests

2.3 Purpose of Data Collection

The collected data provides the foundation for analysing **patient flow, departmental performance, resource utilization, hospital occupancy, and financial activities** through interactive dashboards.

Data Collection Flow

Kaggle → Raw Hospital Dataset → Jupyter Notebook for Refinement

	A	B	C	D	E	F	G	H	I	J	K
1	admission	admission	discharge_admission	admission	patient_id	departmer	ward_id	bed_id	disease_id		
2	1	Emergency	Discharge	166	2	6	76	10			
3	2	Emergency	Discharge	8622	5	21	302	11			
4	3	Emergency	Discharge	23976	1	2	11	9			
5	4	Emergency	Discharge	16635	2	10	128	1			
6	5	Emergency	Discharge	10654	3	11	157	7			
7	6	Emergency	Discharge	21044	3	11	143	14			
8	7	Emergency	Discharge	10798	2	6	79	19			
9	8	Emergency	Discharge	15425	4	19	276	18			
10	9	Emergency	Discharge	13814	1	4	55	16			
11	10	Emergency	Discharge	1654	1	2	11	2			
12	11	Emergency	Discharge	27934	6	25	369	15			
13	12	Emergency	Discharge	21107	5	23	334	11			
14	13	Emergency	Discharge	20420	2	10	140	4			
15	14	Emergency	Discharge	9345	2	7	86	4			
16	15	Emergency	Discharge	21042	3	11	157	16			
17	16	Emergency	Discharge	22916	4	16	240	17			
18	17	Emergency	Discharge	21250	5	23	338	15			
19	18	Emergency	Discharge	25308	5	21	308	16			
20	19	Emergency	Discharge	6947	5	23	343	4			
21	20	Emergency	Discharge	24098	2	10	130	19			
22	21	Emergency	Discharge	8999	4	18	265	20			
23	22	Emergency	Discharge	542	2	10	131	3			
24	23	Emergency	Discharge	8014	4	17	253	12			
25	24	Emergency	Discharge	27939	6	26	390	2			
26	25	Emergency	Discharge	20190	2	8	109	10			
27	26	Emergency	Discharge	10590	1	3	29	12			

Fig:1

3. DATA UNDERSTANDING

3.1 Dataset Structure

The collected hospital dataset contains 45,000 records and 41 columns. Each record represents a hospital admission and contains information about the patient, admission, department, ward, treatment, billing, and payment.

3.2 Major Data Categories

The important fields were grouped into the following categories:

- **Patient Information:** Patient ID, gender, date of birth, blood group, and city.
- **Admission Information:** Admission ID, admission date, discharge date, admission type, and admission status.
- **Hospital Resources:** Department, ward, bed, bed status, and total beds.
- **Medical Information:** Disease, disease category, prescriptions, drugs, diagnostic tests, and doctors.
- **Financial Information:** Total amount, insurance coverage, patient payable amount, room, drug, test, and procedure charges.
- **Payment Information:** Payment status and payment mode.

3.3 Initial Data Checks

Before visualization, the dataset was examined for **missing values, duplicate records, incorrect data types, inconsistent categorical values, and other data-quality issues**. These checks helped determine the preprocessing required before creating the final analytical dataset.

Key Observation

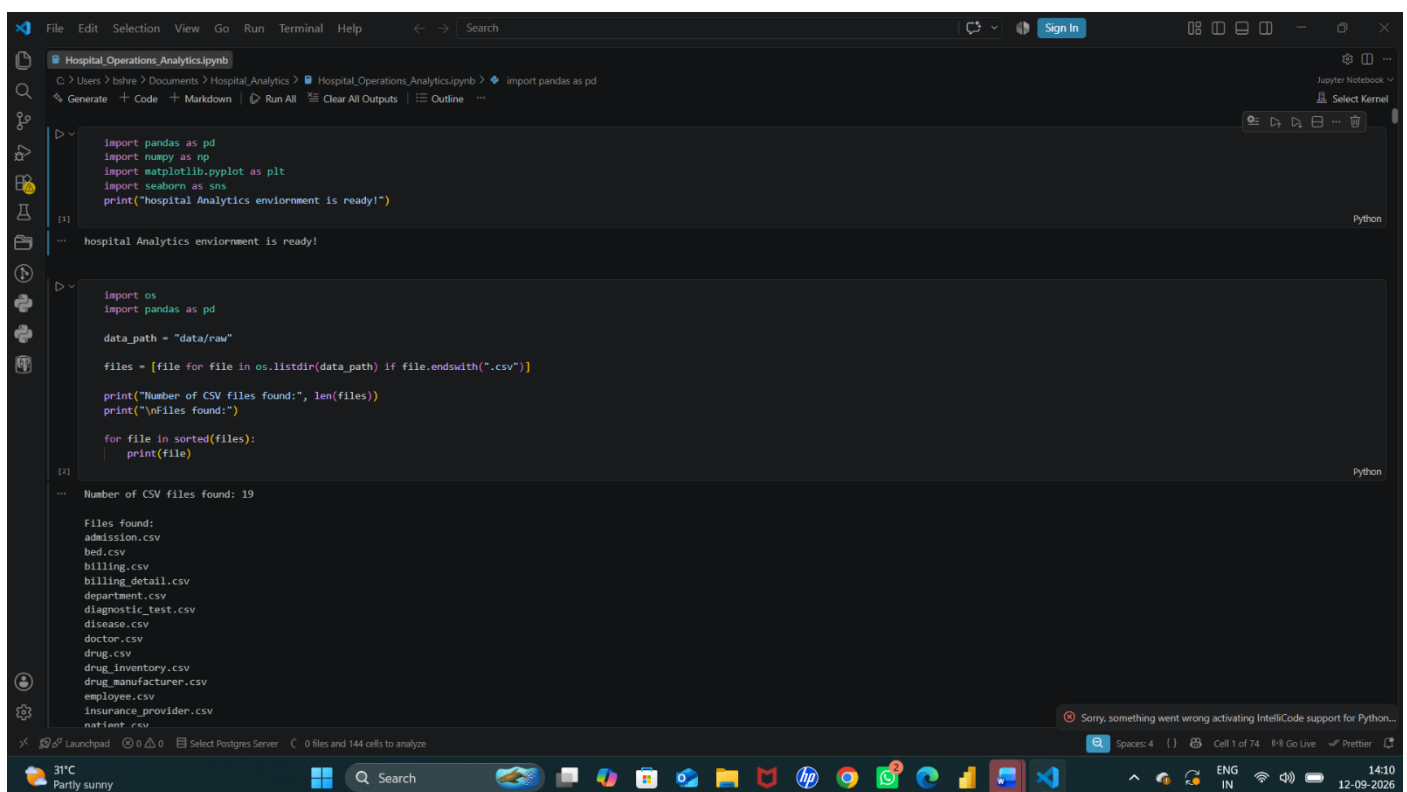
The dataset provided sufficient information to analyse **patient flow, departmental performance, hospital resources, and financial activities**, making it suitable for the planned Tableau dashboards

4.DATA CLEANING & REFINEMENT USING JUPYTER NOTEBOOK

The collected Kaggle dataset was refined using Python and Pandas in Jupyter Notebook before importing it into Tableau. The objective was to improve data quality, maintain consistency, and prepare the dataset for accurate analysis and visualization.

4.1 Loading and Inspecting the Dataset

The raw hospital dataset was imported into Jupyter Notebook using Pandas. Initial inspection was performed to understand the number of records, columns, data types, unique values, and overall structure of the dataset.



```
File Edit Selection View Go Run Terminal Help
Hospital_Operations_Analytics.ipynb
C:\Users> bshree> Documents> Hospital_Analytics> Hospital_Operations_Analytics.ipynb> import pandas as pd
Generate + Code + Markdown Run All Clear All Outputs Outline
[1]
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
print("hospital Analytics enviroenment is ready!")
hospital Analytics enviroenment is ready!
Python

[2]
import os
import pandas as pd

data_path = "data/raw"

files = [file for file in os.listdir(data_path) if file.endswith(".csv")]

print("Number of CSV files found:", len(files))
print("\nFiles found:")

for file in sorted(files):
    print(file)
Number of CSV files found: 19

Files found:
admission.csv
bed.csv
billing.csv
billing_detail.csv
department.csv
diagnostic_test.csv
disease.csv
doctor.csv
drug.csv
drug_inventory.csv
drug_manufacturer.csv
employee.csv
insurance_provider.csv
patient.csv
Sorry, something went wrong activating IntelliCode support for Python..
Launchpad 31°C Partly sunny 0 files and 144 cells to analyze Spaces: 4 Call 1 of 74 ENG IN 14:10 12-09-2026
```

Fig:2

4.2 Checking Data Quality

The dataset was examined for common data-quality issues, including:

- Missing or null values
- Duplicate admission records
- Incorrect or inconsistent data types
- Duplicate or inconsistent categorical values
- Invalid or unexpected values in important fields

4.4 Data Validation

After cleaning, important relationships within the dataset were verified to ensure that the final results were reliable. This included checking **admission counts, patient counts, bed capacity, occupied and available beds, and other important analytical values.**

4.5 Preparing the Final Dataset

After the cleaning and validation process, the refined dataset was exported as:

hospital_cleaned.csv

This cleaned dataset was then used as the primary data source for creating the **four interactive Tableau dashboards.**

Outcome

The preprocessing stage transformed the raw Kaggle data into a **structured, consistent, and analysis-ready dataset**, reducing data-quality issues and ensuring that the KPIs and visualizations developed in Tableau were based on reliable data.

5.TOOLS & TECHNOLOGIES USED

The project used a combination of data collection, data-processing, visualization, and documentation tools.

5.1 Kaggle

Used as the **source of the raw hospital dataset** for the project.

5.2 Python

Used for **data preprocessing, validation, and analysis** during the data-refinement stage.

5.3 Jupyter Notebook

Used as the working environment for **loading, inspecting, cleaning, refining, and validating the hospital dataset**.

5.4 Pandas

Used for handling the dataset in Python, including **data inspection, cleaning, transformation, and validation**.

5.5 Tableau

Used to convert the cleaned dataset into **interactive dashboards, KPIs, charts, filters, and visual analysis**.

5.6 GitHub

Used for **storing, organizing, documenting, and submitting the project work module-wise**.

Overall Technology Workflow

Kaggle → Python/Pandas/Jupyter Notebook → Cleaned CSV → Tableau → GitHub

6.DATA VISUALIZATION USING TABLEAU

After the data was cleaned and saved as hospital_cleaned.csv, it was imported into **Tableau** for visualization and dashboard development.

The visualization stage focused on converting the processed hospital data into **KPIs, charts, filters, and interactive dashboards** that make hospital operations easier to understand.

6.1 Tableau Data Connection

The cleaned CSV file was connected to Tableau as the primary data source. The available patient, admission, department, ward, bed, medical, billing, and payment fields were reviewed before creating the visualizations.

6.2 KPI Development

Important performance indicators were created using Tableau calculations and aggregations, including:

- Total Admissions
- Average Length of Stay
- Readmission Rate
- Occupancy Rate
- Emergency Admission Percentage
- Total and Available Beds
- Department-level performance indicators

These KPIs provide a quick summary of the hospital's operational performance.

6.3 Dashboard Development

Four interactive dashboards were developed:

1. **Hospital Overview** — provides an overall view of admissions, occupancy, patient stay, readmissions, and billing.
2. **Patient Flow** — focuses on admissions, discharges, patient movement, length of stay, and disease categories.
3. **Department Analytics** — compares department workload, revenue, length of stay, admission patterns, and efficiency.

4. **Resource Utilization** — focuses on beds, occupancy, ward capacity, patient distribution, and payment modes.

6.4 Interactive Features

Common filters were added to allow users to dynamically explore the data by:

Admission Type, Department, Gender, Ward Type, and Admission Date.

Dashboard navigation buttons were also added to allow users to move easily between the four dashboards.

6.5 Creating Visualizations

Different chart types were selected according to the nature of the information being analysed:

- **Line Charts** — used for trends over time.
- **Bar/Column Charts** — used to compare categories and departments.
- **Pie Charts** — used to show proportions and distributions.
- **Area Charts** — used to show changes and comparisons over time.
- **Treemaps** — used to represent category-wise volume.
- **Scatter Charts** — used to compare multiple departmental measures.
- **Funnel Charts** — used for comparative distribution of selected measures.
- **Waterfall Chart** — used to represent changes across billing categories.

6.6 Dashboard Design

A consistent visual theme was maintained across all four dashboards using a **dark navy background, contrasting text, colour-coded visualizations, KPI cards, and a common filter panel.**

6.7 Dashboard Navigation

Navigation buttons were incorporated into the dashboards to allow users to move between the four analytical views:

Hospital Overview → Patient Flow → Department Analytics → Resource Utilization

7.HOSPITAL OVERVIEW

The **Hospital Overview** dashboard provides a high-level view of hospital operations by combining patient activity, bed utilization, readmissions, and financial information.

KPIs

1. Total Admissions — 45,000

Represents the total number of hospital admissions recorded in the dataset and provides an overall measure of patient activity.

2. Occupancy Rate — 65.06%

Shows the percentage of hospital beds being utilized, helping assess the current level of resource usage.

3. Average Length of Stay — 5.16 days

Represents the average number of days patients remain hospitalized, providing an indication of patient stay patterns.

4. Readmission Rate — 56.92%

Shows the proportion of patients who were admitted more than once, helping identify repeated hospital utilization.

5. Bed Utilization — 65.06%

Indicates the proportion of total available beds being utilized by patients.

Charts

1. Monthly Admissions Trend — Line Chart

Displays admission volumes month by month and helps identify changes and patterns in patient admissions over time.

2. Readmission Rate Trend — Line Chart

Tracks the readmission rate across months to observe how repeat admissions vary over time.

3. Bed Occupancy — Pie Chart

Compares occupied and available beds, giving a quick visual representation of hospital bed utilization.

4. Admissions by Department — Bar Chart

Compares admission volumes across departments and identifies departments with higher patient workloads.

5. Department Efficiency Comparison — Area Chart

Provides a comparative view of department efficiency and helps identify differences in departmental performance.

6. Patient Age Distribution — Column Chart

Shows the number of patients across different age groups, providing an overview of the hospital's patient demographics.

7. Payment Status Distribution — Pie Chart

Displays the proportion of paid and pending payments recorded in the hospital data.

8. Total Billing Amount by Department — Column Chart

Compares the total billing amount generated by each department.

9. Billing Amount by Payment Status — Waterfall Chart

Shows how the billing amount is distributed across different payment-status categories.

Filters

Admission Type, Department, Gender, Ward Type, and Admission Date are provided on the left side of the dashboard. These interactive filters allow users to customize the dashboard by selecting specific admission categories, departments, patient genders, ward types, or a particular date/date range. The selected filters are applied across the dashboard visuals, allowing users to focus on specific groups and examine their **admission patterns, patient activity, bed utilization, readmissions, and billing information** more closely.

Navigation

A navigation bar is provided at the top of the dashboard with links to **Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization**. It allows users to move easily between the four dashboards and view different aspects of hospital operations without leaving the Tableau presentation.



Fig:3

8.PATIENT FLOW

Patient Flow Dashboard

The **Patient Flow Dashboard** focuses on understanding how patients move through the hospital, from admission to discharge. It provides insights into admission patterns, patient volume, length of stay, gender distribution, and disease categories.

Key Performance Indicators

- **Total Admissions — 45,000:** Shows the total number of hospital admissions.
- **Total Patients — 23,275:** Represents the number of unique patients.
- **Total Discharges — 45,000:** Shows the total number of recorded discharges.
- **Average Length of Stay — 5.16 days:** Indicates the average duration of hospitalization.
- **Emergency Admission — 40.17%:** Shows the proportion of admissions classified as emergency cases.
- **Peak Patient Load:** Highlights the highest observed patient load used in the dashboard.

Visualizations

- **Monthly Admissions vs Discharges — Line Chart:** Compares admission and discharge patterns over time.
- **Average Length of Stay by Department — Ribbon/Area Visualization:** Compares the average stay across departments.
- **Patient Distribution by Gender — Pie Chart:** Shows the proportion of patients by gender.
- **Length of Stay Distribution — Column Chart:** Displays how frequently different lengths of stay occur.
- **Average LOS by Admission Type — Horizontal Bar Chart:** Compares hospitalization duration for different admission types.
- **Patient Movement by Department — Horizontal Bar Chart:** Shows patient volume across departments.
- **Monthly Patient Load — Line Chart:** Tracks unique patient volume over time.
- **Admissions by Disease Category — Treemap:** Represents admission volume across disease categories.
- **Patient Load by Gender Over Time — Area Chart:** Shows changes in patient volume by gender over time.

Filters

The dashboard includes **Admission Type**, **Department**, **Gender**, **Ward Type**, and **Admission Date** filters. These allow users to narrow the analysis and examine patient flow for specific admission categories, departments, genders, ward types, or time periods.

Navigation

The top navigation bar connects the **Patient Flow** dashboard with the other three dashboards—**Hospital Overview**, **Department Analytics**, and **Resource Utilization**—allowing users to switch between different areas of hospital analysis easily.



Fig:4

9.DEPARTMENT ANALYTICS

Department Analytics Dashboard

The **Department Analytics Dashboard** focuses on comparing the performance, patient volume, revenue, length of stay, and admission patterns of different hospital departments. It helps identify high-volume departments and departments with higher or lower operational performance.

Key Performance Indicators

- **Total Departments — 6:** Shows the number of major departments included in the analysis.
- **Average Length of Stay — 5.16 days:** Represents the overall average hospitalization duration.
- **Top Volume Department — Surgery:** Identifies the department with the highest admission volume.
- **Highest LOS Department — ICU — 9.98 days:** Shows the department with the highest average length of stay.
- **Top Efficiency Department:** Highlights the department ranked highest according to the efficiency metric used in the dashboard.

Visualizations

- **Patient Volume by Department — Area Chart:** Compares patient volume across departments.
- **Department Revenue Share — Pie Chart:** Shows each department's contribution to total hospital revenue.
- **Average LOS by Department — Column Chart:** Compares average length of stay between departments.
- **Admission Type Distribution — Donut Chart:** Shows the distribution of elective and emergency admissions.
- **Admission Rate by Department — Horizontal Bar Chart:** Compares admission volumes across departments.
- **Department Comparison — Scatter Plot:** Compares admissions, average LOS, and department efficiency together.
- **Gender Distribution — Column Chart:** Shows patient distribution by gender.

Filters

The dashboard uses **Admission Type, Department, Gender, Ward Type, and Admission Date** filters. These allow users to focus the departmental analysis on particular patient groups, admission categories, departments, ward types, or time periods.

Navigation

The navigation bar at the top connects **Department Analytics** with **Hospital Overview, Patient Flow, and Resource Utilization**, allowing users to move between the four dashboards and compare different areas of hospital performance.



Fig:5

10.RESOURCE UTILIZATION

Resource Utilization Dashboard

The **Resource Utilization Dashboard** focuses on the availability and usage of hospital resources, particularly beds. It helps monitor bed capacity, occupancy, ward-level availability, admission patterns, and payment modes.

Key Performance Indicators

- **Total Beds — 415:** Represents the total bed capacity included in the hospital dataset.
- **Occupied Beds — 270:** Shows the number of beds considered occupied.
- **Available Beds — 145:** Shows the number of beds currently available.
- **Bed Utilization — 65.06%:** Represents the proportion of total bed capacity being utilized.

Visualizations

- **Bed Availability by Ward — Horizontal Bar Chart:** Compares the number of available beds across wards.
- **Occupancy Rate Trend — Line Chart:** Shows changes in bed occupancy over time.
- **Bed Count by Ward — Column Chart:** Compares the total bed capacity of different wards.
- **Admission by Type — Funnel Chart:** Shows the distribution of admissions by admission type.
- **Bed Occupancy by Department — Funnel Chart:** Compares occupied-bed levels across departments.
- **Bed Status Distribution — Pie Chart:** Displays the proportion of occupied and available beds.
- **Patient Distribution by Ward Type — Treemap:** Shows patient volume across different ward types.
- **Payment Mode Distribution — Pie Chart:** Represents the distribution of admissions according to payment mode.

Filters

The dashboard contains **Admission Type, Department, Gender, Ward Type, and Admission Date** filters. These filters allow users to analyze resource utilization for specific departments, patient groups, ward types, admission categories, and selected time periods.

Navigation

The top navigation bar links **Resource Utilization** with the other three dashboards—**Hospital Overview**, **Patient Flow**, and **Department Analytics**. This provides a consistent way to move between the different sections of the MedTrack DV analysis.

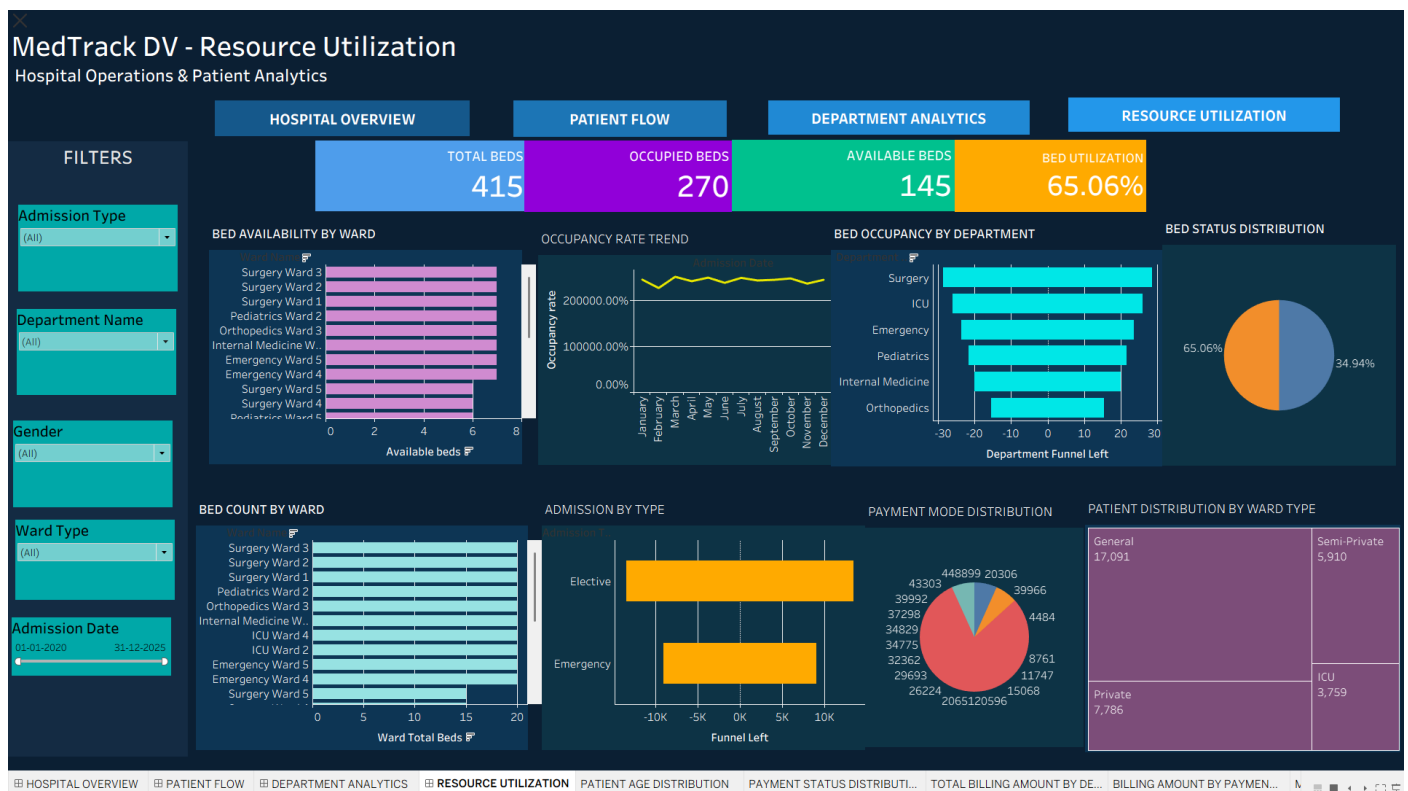


Fig:6

11.DASHBOARD NAVIGATION

Dashboard Navigation

To make the MedTrack DV report easier to explore, a **common navigation bar** was added at the top of all four dashboards. The navigation provides direct access to the major sections of the project:

- **Hospital Overview** — Provides an overall view of hospital operations, admissions, beds, readmissions, and billing.
- **Patient Flow** — Focuses on admissions, discharges, patient movement, length of stay, and patient trends.
- **Department Analytics** — Compares department-wise patient volume, revenue, LOS, admission patterns, and efficiency.
- **Resource Utilization** — Focuses on bed capacity, availability, occupancy, ward utilization, and payment modes.

The navigation buttons were created using **Tableau Dashboard Navigation**, with each button configured to open its corresponding dashboard. This provides a consistent navigation experience throughout the project and allows users to move between dashboards without manually returning to the workbook or dashboard selection screen.

Interactive Filtering

In addition to navigation, the dashboards use common filters such as **Admission Type, Department, Gender, Ward Type, and Admission Date**. These filters allow users to interactively change the displayed analysis and focus on specific categories or time periods.

Navigation Flow

Hospital Overview → Patient Flow → Department Analytics → Resource Utilization

The navigation can also be used to move back to any previous dashboard, making the complete dashboard set easy to explore.



Fig:7

12.KEY FINDINGS

Key Findings

The analysis of the cleaned hospital dataset and the four Tableau dashboards provided several important insights into hospital operations.

- **High Admission Volume:** The dataset contains **45,000 admissions**, providing a substantial basis for analyzing hospital activity and patient flow.
- **Patient Repetition:** The **readmission rate is 56.92%**, indicating that a significant proportion of patients had more than one recorded admission.
- **Bed Utilization:** The hospital has **415 beds**, with **270 occupied** and **145 available**, resulting in a **65.06% bed utilization rate**.
- **Emergency Admissions:** Emergency cases account for approximately **40.17% of total admissions**, showing a considerable demand for urgent hospital services.
- **Average Hospital Stay:** The overall average length of stay is **5.16 days**, while the ICU has a substantially higher average stay of approximately **9.98 days**.
- **Departmental Volume:** **Surgery** has the highest admission volume, with **10,126 admissions**, making it the highest-volume department in the dataset.
- **Payment Patterns:** **Insurance** is the dominant payment mode, accounting for approximately **79.95%** of admissions, while the remaining payments are distributed among UPI, card, and cash.
- **Financial Analysis:** Department-wise billing visualizations help identify differences in revenue contribution and provide a financial perspective on hospital operations.

Overall Insight

The dashboards bring together **patient flow, departmental performance, resource utilization, and financial information** into one interactive analytical system. This makes it easier to identify operational patterns and compare different areas of hospital performance.

13. CHALLENGES & SOLUTIONS

Challenges & Solutions

During the development of the MedTrack DV project, several challenges were encountered while working with the dataset, data-cleaning process, and Tableau dashboards.

Challenge	Solution
Raw dataset required refinement	The dataset was inspected and cleaned using Python and Pandas in Jupyter Notebook before visualization.
Inconsistent or unsuitable data fields	Important fields were checked for data types, unique values, duplicates, and consistency before being used in Tableau.
Creating calculated KPIs	Tableau calculated fields and appropriate aggregations such as COUNTD, AVG, and SUM were used to create the required KPIs.
Representing certain Power BI visuals in Tableau	Alternative Tableau techniques such as Gantt-based funnels and area-chart approximations were used where an exact native visual was unavailable.
Maintaining dashboard consistency	A common dark-theme design, filter panel, KPI style, and navigation bar were maintained across all four dashboards.
Making dashboards interactive	Filters and dashboard navigation were configured so users could explore different departments, admission types, genders, ward types, and dates.

Overall Challenge

One of the main challenges was **translating the reference dashboard designs into Tableau while maintaining accurate values from the cleaned dataset**. The project required both data-analysis understanding and practical knowledge of Tableau's visualization and dashboard features.

Solution Approach

The challenges were addressed through **data validation, calculated fields, appropriate aggregations, Tableau formatting, and repeated testing of dashboard visuals**. This helped ensure that the final dashboards were both interactive and based on the refined hospital dataset.

14.CONCLUSION

The **MedTrack DV – Hospital Operations & Patient Analytics** project successfully demonstrates the complete process of transforming raw hospital data into meaningful and interactive analytical dashboards. The project began with the collection of a hospital dataset from **Kaggle**, which was then studied to understand its structure, attributes, and relevance to hospital operations.

The raw dataset was refined using **Python, Pandas, and Jupyter Notebook**. Data cleaning and validation were important steps in ensuring that the dataset was consistent, structured, and suitable for analysis. Different aspects such as duplicate records, data types, categorical values, numerical fields, admission information, patient details, hospital resources, and financial information were reviewed during the refinement process. The final cleaned dataset was then used as the foundation for visualization.

Using **Tableau**, four interactive dashboards were developed: **Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization**. Each dashboard was designed to address a different aspect of hospital operations. The dashboards use a combination of **KPI cards, line charts, bar charts, area charts, pie charts, treemaps, scatter plots, funnel-style visualizations, and waterfall visualizations** to present information in an easily understandable format.

The project also provided practical experience in creating **calculated fields, aggregations, interactive filters, dashboard layouts, navigation buttons, and consistent visual themes**. The common navigation system allows users to move between the four dashboards, while filters provide the ability to explore the data based on admission type, department, gender, ward type, and admission date.

The analysis provided useful insights into hospital operations, including **45,000 admissions, 23,275 unique patients, a 5.16-day average length of stay, 65.06% bed utilization, and a 40.17% emergency admission rate**. Department-level analysis also highlighted differences in patient volume and length of stay, while resource and payment visualizations provided additional operational and financial perspectives.

Overall, this project helped demonstrate how **data analysis and data visualization can convert complex healthcare data into clear, interactive, and useful information**. It also provided practical experience in working through the complete data analytics workflow—from **data collection and cleaning to visualization, dashboard development, analysis, and presentation**. The MedTrack DV dashboards therefore serve as a consolidated analytical view of hospital operations and demonstrate the value of visualization in understanding patterns and supporting data-driven decision-making.

15.REFERENCES

References

The following resources and tools were used during the development of the **MedTrack DV – Hospital Operations & Patient Analytics** project:

1. **Kaggle** — Used as the source for the raw hospital dataset.
2. **Python** — Used for data preprocessing, cleaning, and validation.
3. **Pandas Documentation** — Used as a reference for data manipulation and analysis.
4. **Jupyter Notebook** — Used for dataset inspection, cleaning, refinement, and validation.
5. **Tableau** — Used for data visualization, calculated fields, KPI creation, interactive filters, dashboard design, and navigation.
6. **Tableau Documentation** — Used as a reference for dashboard, worksheet, calculated-field, and visualization features.
7. **GitHub** — Used for organizing, storing, and submitting the project work module-wise.

Project Resources

- Raw hospital dataset obtained from **Kaggle**
- Refined dataset: **hospital_cleaned.csv**
- Jupyter Notebook used for data cleaning and validation
- Tableau workbook containing the four MedTrack DV dashboards
- Dashboard reference/storyboard used for maintaining the required visual structure and design